

# Jahnavi Dulala

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## Professional Summary

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**Machine Learning Engineer / Data Scientist** with production **ML** experience at **Bosch Rexroth** and prior data analytics experience at **Amazon**. **M.Sc. Mechatronics**, Universität Siegen, with coursework and thesis focused on **machine learning** and **sensor analytics**. Builds end-to-end systems rather than notebooks: **ETL** and **feature pipelines**, distributed training on **AWS SageMaker**, containerised deployment with **Docker** and **Airflow**, and **MLflow experiment tracking**. Comfortable across the stack from embedded signal compression to **Retrieval-Augmented Generation**.

## Technical Skills

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**Programming & Core:** Python (NumPy, Pandas, SciPy), **SQL** (PostgreSQL, BigQuery), C++, MATLAB, Bash, Git

**Machine Learning:** scikit-learn, XGBoost, Random Forest, Feature Engineering, Anomaly Detection, Recommender Systems (Implicit, LensKit), Explainable AI (SHAP, LIME)

**Deep Learning & GenAI:** PyTorch, TensorFlow, Keras, CNN, LSTM, Autoencoders, Hugging Face, Transformers, Retrieval-Augmented Generation (RAG), FAISS, OpenCV

**MLOps & Cloud:** Docker, Apache Airflow, MLflow, AWS SageMaker, S3, Google Cloud Platform (BigQuery, Vertex AI), FastAPI, REST APIs, CI/CD, Optuna, Grafana

**Analytics & Visualisation:** Streamlit, Power BI, Tableau, Plotly, Matplotlib, Seaborn, Excel

**Languages:** English (C2-fluent), German (A2/B1-intermediate, actively improving)

## Professional Experience

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### Master Thesis Researcher

Dec 2024 – Sep 2025

Universität Siegen

Siegen, Germany

- Designed a **Python/JavaScript quantisation pipeline** benchmarking four compression methods for 12-channel wearable sensor data, cutting payload size by **97%** (20 MB to under 1 MB).
- Established the practical **accuracy-compression boundary** for on-device **edge deployment** by measuring downstream **Human Activity Recognition (Tiny-HAR)** performance across compression levels, defining the usable operating range for lossy sensor encoding.
- Implemented **1-byte-per-sample encoding** on the **WEAR benchmark dataset** with temporal activity patterns preserved.
- Delivered a **browser-based visualisation tool** rendering at **100 FPS** with **sub-50 ms decode latency** for interactive time-series exploration.

### Machine Learning Intern

Jun 2024 – Nov 2024

Bosch Rexroth

Ulm, Germany

- Built an end-to-end **fault-detection system** for production hydraulic sensors, covering **ETL**, signal feature extraction, and deployment, replacing manual inspection with continuous real-time **anomaly detection**.
- Benchmarked **Linear Autoencoder**, **CNN**, and **LSTM-Autoencoder** architectures using distributed **AWS SageMaker** training jobs with **Optuna hyperparameter search** to select the production model.
- Industrialised the team's **MLOps workflow**: containerised training with **Docker**, orchestrated it in **Apache Airflow**, and added **MLflow experiment logging** and **Grafana health telemetry**, removing manual run tracking.
- Shipped interactive **Explainable AI dashboards** (SHAP, LIME) in **Streamlit**, giving domain engineers model interpretability and real-time drift visibility without ML expertise.

### Student Researcher - Recommender Systems

Aug 2023 – Apr 2024

Universität Siegen

Siegen, Germany

- Ran **96+ controlled hypothesis experiments** on **MovieLens**, quantifying how top-percentile user pruning affects **Implicit ALS** versus **KNN recommenders**.
- Identified a structural asymmetry between the two model families at equivalent sparsity thresholds: a **10.67% NDCG penalty** for ALS against a **9.6% NDCG gain** for KNN.
- Automated the **evaluation pipeline** (LensKit, Implicit, scikit-learn, Seaborn) to produce reproducible comparative statistics, cutting turnaround between experiment rounds.

## Business Operations Analyst (Data Analyst)

Amazon

Mar 2020 – Jul 2021

Hyderabad, India

- Raised catalogue defect-identification precision by **11%** by redesigning **SQL preprocessing pipelines** and aligning **ETL validation rules** with SOP protocols.
- Traced root-cause drivers of customer complaints and counterfeit listings through cross-tabulated **anomaly analysis** in **SQL, Power BI, and Excel**, reducing operational escalations.
- Architected automated **Power BI KPI dashboards** that became the cross-functional reference for weekly quality reporting.
- Led **SQL tuning** and data-tagging workshops for **10+ junior analysts**, shortening team onboarding cycles.

## Projects

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### Enterprise Document Intelligence Assistant (RAG / NLP)

Python, FAISS, Transformers, Gradio

- Architected a **retrieval-augmented generation pipeline** over a **15K-article BBC News corpus** using **FAISS vector indexing** and **Sentence Transformer embeddings**, grounding every answer in retrieved sources to eliminate hallucinated responses; served through an interactive **Gradio UI**.

### Image Denoising & Restoration Dashboard

PyTorch, U-Net, Gradio, Hugging Face

- Deployed a **Hugging Face Spaces application** comparing five deep learning restoration models (**CBDNet**, **IRCNN**, **U-Net**, **RCAN**, Autoencoders) across six **no-reference perceptual quality benchmarks**.

## Education

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### M.Sc. Mechatronics

Universität Siegen

Oct 2021 – Sep 2025

Siegen, Germany

- Focus: Machine Learning, Deep Learning, Signal Processing, Embedded Control, Project Management.

### B.E. Mechanical Engineering

Vardhaman College of Engineering

Oct 2015 – Jun 2019

Hyderabad, India

- Focus: Auto CAD, Material Science, Engineering Electronics, Entrepreneurship, Hydraulics.

## Certifications

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- **Agentic AI on Google Cloud** - Google (Jul 2026)
- **Introduction to AI and Machine Learning on Google Cloud** - Google (Feb 2026)
- **BigQuery for Data Analysts** - Google (Nov 2025)
- **Machine Learning Specialization** - DeepLearning.AI / Coursera (Jun 2025)
- **Generative AI Fundamentals** - Databricks (Oct 2025)